

5A-317-250



Precision Industries
Testing Department

GENERATING-SET

LOAD TEST

Model No: PI 250 C 60 Hz

Type: Close Job No: 9428

Color: White Date: 12/08/2018

Serial Number:								
PI Model No:	PI 250 C	Prime Power:	225	[kVA]	P.F (Load):	1	Voltage:	220

Generating Set Details:

Engine:	Cummins	Model No: QSB7 - G5	Serial No: 84468619
Alternator:	Stamford	Model No: UCI 274 H1	Serial No: N17D176584
Controller:	DSE	Model No: 6120	Serial No: 6754505
C. Breaker:	Metasol 800 A	Model No: ABN 803 C	Serial No: N/A
C.Transformer:	Ziegler 800/5A	Model No: N/A	Serial No: N/A
AVR:	Stamford	Model No: SX 460	Serial No: 111505 - 2014
Cooling System: Water Cool	Enclosure Type: Canopy	Protection Class: IP23	

Load Test Ambient Conditions:

Temperature	38	[°C]	Altitude:	0	[m]	Humidity:	53	%
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Generating Set Rating:

Testing Full Load Current:	472.38	Amps	Nominal Current:	590.5	Amps
Power Correction Factor (De-rating):	0.000	%	Corrected Prime Power:	0.0	[kVA]

LOAD TEST

Load	Run Time	Freq	Speed	VOLTAGE L-L			CURRENT			Power	Coolant Temp	Oil Pressure	Charg-Alt.
				L1	L2	L3	L1	L2	L3				
%	min	Hz	RPM	V	V	V	A	A	A	kW	C	bar	VDC
0 %	5	60	1800	220	220	220	-	-	-	-	73	4.68	14.6
25 %	5	60	1800	220	220	220	118	117	117	45.0	81	4.28	14.6
50 %	5	60	1800	220	220	220	239	238	238	91.0	83	3.96	14.6
75 %	5	60	1800	220	220	220	356	356	355	135.0	85	3.8	14.8
100 %	10	60	1800	220	220	220	474	473	472	181.0	89	3.48	14.8
110 %	10	60	1800	220	220	221	519	519	518	198.0	95	3.32	14.8

LOAD ACCEPTANCE TEST

Suddenly Block Load %	Voltage [V]	Frequency [Hz]	Current [A]	Response Time [sec]
50%	220	60.0	239	0
0%	227 - 220	61.8 - 60.0	0	2
50%	216 - 220	58.4 - 60.0	242	2

PROTECTION SYSTEM TEST

Over Frequency Warning / SD	Under Frequency Warning / SD	Over Voltage Warning / SD	Under Voltage Warning / SD	Oil Pressure SD	Coolant Level Warning / SD	Coolant Temp Warning / SD
65 / 67	55 / 57	240 / 260	190 / 180	1.0	OK	103 / 108 c

ACCESSORIES DETAILS

Model:		Serial No.	
Model:		Serial No.	

[Signature]

PRECISION INDUSTRIES
TESTED
Testing Department

PI Document no. PI-TST-LT-01 Rev. 01

Test Certificate

FOR

SALIENT POLE, SELF EXCITED / PMG EXCITED AND SELF REGULATED AC GENERATOR

Machine No. **N17D176584**

Type Test has been conducted on similar design machine
STANDARDS: Generator generally conforms to BS 5000 : Part 3,
 Tested in accordance with BS EN 60034-1/IS/IEC 60034-1
 Rotor dynamically balanced to BS ISO 1940 - 1

ROUTINE TEST CERTIFICATE FOR BRUSHLESS AC GENERATOR:

FRAME: UCI274H1	AVR: SX460	AVR NO: 1115052014	ENCLOSURE: IP23	
RATED KVA: 200	RATED VOLTS: 400	RATED AMPS: 288.7	FREQUENCY: 50	PHASE: 3
POWER FACTOR: .8	STR CONN: Series Star	STR WINDING: 311	DUTY: Base Continuous	WIRE: 4-Wire
EXC. VOLTS: 58	EXC.AMPS: 2.3	AMB TEMP: 40		

FOR 3 PHASE SEQUENCE LOOKING FROM DRIVE END FOR CLOCKWISE ROTATION :U-V-W

WINDING	COLD RESISTANCE IN Ohm(+/- 10 Milli Ohm) AT 20°C (MAX VALUES)	INSULATION RESISTANCE (M- Ohm)	HV TEST 2KV FOR 1 MIN	INSULATION CLASS
STATOR	0.03371	2+	OK	H
ROTOR	1.9663	2+	OK	H

REGULATION TEST

FREQ.	VOLTS	AMPS	KVA	PF	KW	EXC.VOLT	EXC.AMP	% LOAD
52	401.35	0.00	0.00	0.00	0.00	12	.5	0.00
50	400	288.7	200	.8	160	58	2.3	100.00

REGULATION = (N.L.VOLTAGE-F.L.VOLTAGE / RATED VOLTAGE)*100 = 0.34 %

Item Description: AC Generator.UC27.4.H.1.311.,3.,,11.5.,.SX460.Black - RAL9011

Remarks:

Date: 26-APR-17	Job No: 3481896-007	AUTHORISED BY Mahesh R Tarde Quality Assurance
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Certificate Of Conformity

Quality System Approval
 ISO 9001/2008

Frame Size : **UCI274H1**Serial Number : **N17D176584**

Manufactured Date : 26-APR-17

KVA: **200**Authorized By Quality Assurance: **Mahesh R Tarde**

EU DECLARATION OF CONFORMITY



**Generator
Technologies**

This synchronous A.C. generator is designed for incorporation into an electricity generating-set and fulfils all the relevant provisions of the following EU Directive(s) when installed in accordance with the installation instructions contained in the product documentation:

2014/35/EU

Low Voltage Directive

2014/30/EU

The Electromagnetic Compatibility (EMC) Directive

and that the standards and/or technical specifications referenced below have been applied:

EN 61000-6-2:2005

Electromagnetic compatibility (EMC). Generic standards - Part 6-2:
Immunity for industrial environments

EN 61000-6-4:2007+A1:2011

Electromagnetic compatibility (EMC). Generic standards - Part 6-4:
Emission standard for industrial environments

EN ISO 12100:2010

Safety of machinery - General principles for design - Risk assessment
and risk reduction

EN 60034-1:2010

Rotating electrical machines - Part 1: Rating and performance

BS ISO 8528-3:2005

Reciprocating internal combustion engine driven alternating current
generating sets - Part 3: Alternating current generators for generating sets

BS 5000-3:2006

Rotating electrical machines of particular types or for particular
applications - Part 3: Generators to be driven by reciprocating internal
combustion engines - Requirements for resistance to vibration

This declaration has been issued under the sole responsibility of the manufacturer. The object of this Declaration is in conformity with the relevant Union harmonization Legislation.

The name and address of authorised representative, authorised to compile the relevant technical documentation, is the Company Secretary, Cummins Generator Technologies Limited, 49/51 Gresham Road, Staines, Middlesex, TW18 2BD, U.K.

Signed:

Date: 01st February 2016

Name, Title and Address:

Kevan J Simon
Global Technical and Quality Director
Cummins Generator Technologies
Fountain Court
Lynch Wood
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PE2 6FZ

Description UCI274H1

Serial Number N17D176584

450-16383-F